JavaScript

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Sr. No	Topic	JavaScript	Python
1	Language Properties	Open Source Object Oriented Event Programming Async I/0 Dynamically Typed	Open Source Object Oriented Dynamically Typed
2	Code comment	// content OR /* content */	# content OR ''' content''' OR """ content"""
3	Print to console	console.log(content)	print(content)
4	Data Types	Number, String, Boolean (**true, false**)	Number (int, float, complex) String Boolean (**True, False**) List, tuple Set, frozen set Dictionary
5	Variable	Var - could be re-declared - variable re-assignment possible - e.g. var x = 12, x = 24 - Var x = 'javascript prog' let - could not be re-declared, - variable re-assignment possible - e.g. let x = 12, x = 24 - ~~let x = 'javascript prog'~~ const - could not be re-declared - variable re-assignment not possible - e.g. const x = 12 - ~~const = 24~~ - ~~let const = 'javascript prog'~~	Dynamically typed Variable re-assignment possible No use of any identifier like var, let, const Simply X = 12 X = 14 X = "python programing"
6	Check data type	typeof(variable)	type(variable)

7	If else	If(condition)
		{
		statement
		}
		else if
		{
		statement
		}
		else
		{
		statement
		}
			If condition:
			Statement
			elif condition:
			Statement
			else:
			Statement
			
			If condition : Statement
			
			Statement if condition else Statement
			
			Statement elif condition if condition else Statemen

8	while	i=0
		while(i<=10){
		i++
		Console.log(i)
		}
			i=0
			while i<=10:
			i++
			print(i)

9	do while	do{ i=0 Console.log(i) i++	do while not available with python

		}while(i<10)**;** Console.log(i)	
10	Array	declaration w/o initialization let x = array(6) declaration with initialization let y = new array([1,2,3,4,5,6,7,8,9]) direct initialization var x = [1,2,3,4,5,6,7,8,9]	x = list() x = [1,2,3,4,5,6,7,8,9]

11	Array Methods	var x = [1,2,3,4,5,6,7,8,9]
		
		console.log(typeof(x)) // array
		
		x.length() // length of array
		
		console.log(x[3]) // 4
		x[3] = 12 // modifying specific index value
		
		x.**push**(12) // to add item at last index
		x.pop() // to remove item at last index
		
		x.**unshift**(14) // to **add** item at **first index**
		x.**shift**() // to **remove** item at **first index**
		x.**splice**(1,2, 23, 24) // to **add** item at **specific index**
		
		x.**indexOf**(item) // return index of item
		x.**includes**(item) // boolean true/flase
		
		num = [1,2,3], car = ['audi', 'bmw', 'mercedes']
		
		y = num.**concat**(car) // **combines two arrays**
		console.log(y)
		
		numbers = [3,6,9,12,15,18,12,15,21,12]
		
		x = **numbers.slice(3,6)** // returns elements from index 3 to 5
			x = [1,2,3,4,5,6,7,8,9]
			
			print(type(x)) # <class List>
			
			length(x) # length of list
			
			print(x[3]) # 4
			x[3] = 12 # modifying specific index value
			
			x.**append**(12) # to add item at last index
			x.pop() # to remove item at last index
			
			x.**insert**(index, item) # insert at specific index
			x.remove(item) # remove specific item
			
			x.index(item) # return index of item
			Item **in** x # boolean return value
			
			num = [1,2,3], car = ['audi', 'bmw', 'mercedes']
			
			y = num.**extend**(car) # **combines two lists**
			console.log(y)
			
			numbers = [3, 6, 9, 12, 15, 18, 12, 15, 21, 12]
			
			x = **numbers[3:6]** # Python uses [start:stop] syntax for slicing

12	Logical operator	&& || !	and, or, not

13	for loop with index	array = [2,4,6,8,10,12]
		
		for(i=0 ; i<array.length ; i++){
		console.log(i)
		}
			even_no_list = [2,4,6,8,10,12]
			
			for i in range(length(even_no_list)):
			    Print(even_no_list[i])

14	For loop with 'in' operator	array = [2,4,6,8,10,12]
		
		for(number **in** array)
		{
		console.log(array[number])
		}
			even_no_list = [2,4,6,8,10,12]
			
			for item **in** even_no_list:
			    Print(item)

15	For loop with 'of' operator	array = [2,4,6,8,10,12]
		
		for(number **of** array)
		{
		console.log(number)
		}
			NA

16	forEach()	fruits = ['mango', 'apple', 'kiwi', 'berry']	fruits = ['mango', 'apple', 'kiwi', 'berry']

		fruits.forEach(fruit => console.log(fruit));
		
		fruits.forEach((fruit, index) => {
		  console.log(`Index: ${index}, Fruit: ${fruit}`);
		});
		
		array = [2,4,6,8,10,12]
		
		array .forEach(num => num*2)
			for fruit **in** fruits:
			        Print(fruit)
			
			array = [2,4,6,8,10,12]
			
			double_number = [num*2 for num in array]

17	find	let numArray = [3,6,9,12,15,18,12,15,21,12] **let firstGreater = numArray.find((num) => num > 10)** console.log(firstGreater)	num_list = [3, 6, 9, 12, 15, 18, 12, 15, 21, 12] **first_greater = next((num for num in num if num > 10), None)** print(first_greater) print(next(first_greater)) # **next()** gets the first item that matches the condition *generator expression inside the next() function, mimics JavaScript's find() method perfectly.*
18	filter	let numArray = [3,6,9,12,15,18,12,15,21,12] **let evenNumbers = numArray.filter((num) => num % 2== 0)** console.log(evenNumbers)	num_list = [3, 6, 9, 12, 15, 18, 12, 15, 21, 12] **even_numbers = list(filter(lambda num: num % 2 == 0, num_list))** ***even_numbers = [num for num in num_list if num % 2 == 0]*** print(even_numbers)
19	map	let numArray = [3,6,9,12,15,18,12,15,21,12] **let doubledNumbers = numArray.map((num) => num*2)** console.log(doubledNumbers)	num_list = [3, 6, 9, 12, 15, 18, 12, 15, 21, 12] **doubled_numbers = list(map(lambda num: num*2, num_list)** ***doubled_numbers = [num * 2 for num in num_array]*** print(doubled_numbers)

20	reduce	let numArray = [3,6,9,12,15,18,12,15,21,12]
		
		sum = 0
		
		for (let i=0; i<numArray.length; i++) {
		  sum += numArray[i]
		}
		console.log(sum)
		
		var total = numArray.reduce((sum,num) => sum+num,0)
		
		console.log(total)
			num_list = [3,6,9,12,15,18,12,15,21,12]
			
			total = sum([num for num in num_list])
			print(total)
			
			OR
			
			total = sum(num for num in num_array)
			
			OR
			
			Sum(num_list)

21	Sort string	fruits = ['mango', 'apple', 'kiwi', 'berry'] console.log(fruits.sort()) console.log(fruits.reverse())	fruits = ['mango', 'apple', 'kiwi', 'berry'] print(fruits.**sort**()) # modifies the original list directly Print(**sorted**(fruits)) #creates a **new sorted list** print(fruits.**reverse**()) # reverse the original list directly print(**reversed**(fruits)) # #creates a **new reversed list**
22	Sort number Bubble sort	numbers = [3,6,9,12,15,18,12,15,21,12] numbers.sort((a,b) => a-b) console.log(numbers) numbers.sort((a,b) => b-a) console.log(numbers)	numbers.sort()

23	Sort	const products = [
		  { name: "Mouse", price: 25 },
		  { name: "Laptop", price: 1200 },
		  { name: "Keyboard", price: 75 },
		  { name: "Monitor", price: 300 }
		];
		
		// In-place sort using an arrow function comparator (a, b)
		
		products.sort((a, b) => a.price - b.price);
		
		console.log(products);
		
		/* Output:
		[
		  { name: 'Mouse', price: 25 },
		  { name: 'Keyboard', price: 75 },
		  { name: 'Monitor', price: 300 },
		  { name: 'Laptop', price: 1200 }
		]
		*/
			products = [
			    {"name": "Mouse", "price": 25},
			    {"name": "Laptop", "price": 1200},
			    {"name": "Keyboard", "price": 75},
			    {"name": "Monitor", "price": 300}
			]
			
			# Creates a new sorted list using a lambda key extractor
			
			sorted_products = sorted(products, key=lambda item: item["price"])
			
			print(sorted_products)
			
			""" Output:
			[
			    {'name': 'Mouse', 'price': 25},
			    {'name': 'Keyboard', 'price': 75},
			    {'name': 'Monitor', 'price': 300},
			    {'name': 'Laptop', 'price': 1200}
			]
			"""